Chapter 3
Students' Entrepreneurship Development Through Indigenous Palm Wine Production and Alcohol Distillation:
The Case of Yala-Obubra Community

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ABSTRACT

This chapter used a qualitative approach to investigate and describe the palm wine production and alcohol distillation practices of the indigenous people of the Yala-Obubra community in Cross River State, Nigeria. The chapter involved palm wine tappers and alcohol distillers (n = 10). A focus group discussion was held with the respondents, with audio recordings and notes for data collection. The study documented several practices in palm wine production and alcohol distillation and discovered declining palm tree population, climate change, high cost of production, lack of access to credit and social amenities, and stealing as some challenges faced in producing palm wine and distillation of alcohol. The chapter recorded some economic opportunities in the indigenous production of palm wine and distillation of alcohol, and discussed the relevance of these two traditional practices in promoting entrepreneurship among students. The implications of this study's findings were discussed for the promotion of sustainable production and preservation of these traditional practices in Nigeria.

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INTRODUCTION AND BACKGROUND

Over the years, there has been a persistent cry in Nigeria among stakeholders due to the issue of unemployment among youths. A significant portion of the unemployed people have graduated from different tertiary institutions in Nigeria (Owan et al., 2022a). In the past, different scholars have advocated for restructuring the tertiary education curriculum to improve students' entrepreneurial skills for job creation (Eje et al., 2021; Owan et al., 2021; 2022b). However, the unemployment situation keeps worsening with each passing year. Recently, Oseni and Oyelade (2023) analysed available World Bank data on unemployment from 1981 to 2020 and revealed that the average proportion of the labour force in Nigeria was 76.77%, with the lowest and highest values falling between 71.7% and 79.35% respectively. In contrast, the rate of employment growth was only 17.45% annually. This substantial difference explains why the compensation for employees is low, with a growth rate of only 35.96%. At the time of writing, Nigeria's unemployment rate stood at 41% (Osabohien, 2023).

The high unemployment rate has created a need for university students to devise new strategies to generate income and create self-employment opportunities. It has been argued that no government can employ all its citizens (Ekaette et al., 2019; Odigwe & Owan, 2019), although highly developed and industrialised nations have small unemployment rates. One way that Nigerian students and graduates can leverage to free themselves from the burden of unemployment and poverty is to consider developing entrepreneurial skills and mindset. Some studies have shown that the entrepreneurship activities of students are critical in poverty reduction (Lee & Rodríguez-Pose, 2021); however, students' behavioural changes were required (Si et al., 2015).

From the preceding, it is imperative for students to consider venturing into different initiatives to enhance their earnings and liberate themselves from poverty. This qualitative research paper discusses the importance of indigenous practices such as local palm wine production and alcohol distillation in enhancing students' entrepreneurship development. Palm wine is a traditional beverage very common in Nigeria, like most African countries that grow palm (Akinrotoye, 2014). It is a whitish liquid consumed throughout the tropics (Uzogara et al., 1990). However, it must be noted that fermented palm wine can produce spirits through a distillation process. The distillation process involves heating the fermented palm wine to a high temperature to produce steam, which is then condensed into liquid form, resulting in a more concentrated and potent alcoholic beverage. Although both can be termed alcoholic drinks, palm wine is different from the spirit produced from it in several regards. First, palm wine is a mildly alcoholic beverage, with an alcohol content ranging from 2% to 6% (Lal et al., 2003). In contrast, the alcohol produced from palm wine is much stronger, with an alcohol content of around 40% to 43% (Tagba et al., 2018). Palm wine has a sweet, slightly sour taste, with a fruity aroma that is often compared to that of champagne. Unfermented palm wine is clean, sweet, colourless syrup, which is mainly sucrose (Okafor, 1975). The alcohol produced from palm wine, however, has a more intense flavour and a stronger, more pungent aroma. Palm wine is produced by tapping the sap of palm trees and collecting it in containers. The alcohol produced from palm wine is made by fermenting it and then distilling it. The alcohol from palm wine is durable and can be bottled for export to other countries. On the contrary, palm wine is a perishable product typically consumed locally and regionally.

Nevertheless, through the pasteurisation process (boiling fresh palm wine up boiling point of 70 to 75^{o}C) followed by refrigeration, palm wine can have a long-lasting duration (Ikegwu, 2014). Pasteurising palm wine at 75°C for 45 minutes and packaging it for up to 24 months with no preservatives does not affect the nutrients or deteriorate the content (Obahiagbon & Oviasogie, 2007). This suggests that pasteurised palm wine and distilled alcohol can be good sources of income for those practising the trade, particularly in rural areas where formal employment opportunities are limited.

Despite the importance of traditional palm wine production and alcohol distillation practices, which have the potential for entrepreneurship development in Africa, these practices are often overlooked and undervalued by policymakers, investors, and other stakeholders. This is partly due to the perception that such traditional practices are obsolete and that modernisation and formalisation are necessary for economic growth and development. Unfortunately, this perspective overlooks the potential of these two traditional practices to drive entrepreneurship and economic growth in Africa, particularly when combined with modern techniques and technologies. For instance, some youths see such practices as occupations for poor or hopeless individuals. Others tend to believe that these practices are meant for those in rural localities. The high rate of rural-urban migration in Nigeria due to the quest for white-collar jobs and employment opportunities (Olorunfemi, 2023) has further limited youth participation in agriculture-related activities such as palm wine production and distillation (Ovharhe et al., 2022).

Producing palm wine and distilled spirits requires complex and expensive scientific procedures. This involves selecting raw materials, milling and mashing to create a fermentable mash, fermenting with yeast, and distilling to separate alcohol from water and impurities. Thus, different devices and machines will be required for production. These might be very expensive for a beginner to afford. However, it is common knowledge that there are traditional approaches to palm wine and distilled spirits production that are less expensive to follow. Nevertheless, the indigenous processes involved in palm wine production and alcohol distillation is not easily accessible since only very few people possess such information. Moreover, the knowledge of these practices cannot be easily accessed from the existing literature or the Internet since such techniques have not been previously documented. Therefore, it is important to extract such information from the few individuals possessing them and document them electronically to guide against its extinction and pass it on to newer generations interested in it. Along these lines, this chapter aims to provide a comprehensive overview of indigenous palm wine production and alcohol distillation practices in Africa and to explore the implications of these practices for entrepreneurship development.

Research Objectives

The objectives of this chapter are to:

- Identify the steps and materials required to tap palm wine in the context of the Yala-Obubra Community;
- Reveal the causes of poor taste in the palm wine of some tappers in the community;
- Uncover the reasons for differentiated drunkenness patterns among different palm-wine consumers;
- Identify the materials and processes involved in indigenous alcohol distillation practices in the community;
- Reveal the challenges facing traditional palm-wine production and distillation of alcohol in the community.

LITERATURE REVIEW

Empirical studies continue to explore ways in which the indigenous palm wine or the yeast associated with it can be used for the production of other useful substances such as vinegar, juice and medicine (Akinrotoye, 2014; Samuel et al., 2016). This interest has been driven by several factors, including the recognition of the potential of indigenous practices to create jobs, generate income and support local development, the increasing demand for high-quality and authentic African products in both local and international markets and the growing awareness of the need to preserve and promote African cultural heritage (Owan et al., 2020).

Studies on entrepreneurship has been conducted to identify ways in which palm wine and alcohol distillation can improve the livelihoods of rural individuals. For instance, Scholtes (2009) found that small-scale ethanol production shows promise for supplying energy in remote areas and establishing related value chains, particularly in rural Africa, for purposes other than transportation. Given the easy integration of sugarcane cultivation and ethanol production through small-scale distilleries, smallholders engaged in cane production can benefit significantly from this approach. Similarly, McCoy et al. (2013) discovered that alcohol production serves as an adaptive livelihood strategy for women in agriculture, allowing them to maintain control over income and support their households, but it also exposes them to increased vulnerability through risky sexual behaviours). Another study discovered that palm wine is a major source of livelihood across the various rural communities studied (Macie & Martins, 2019). Studies on entrepreneurship have shown that income can be generated from numerous sources. For example, Odigwe et al. (2018) identified baking and computer business operations as two sources of income generation for students. Another study identified farm enterprises as a sustainable source of vocational venture for students (Francis et al., 2019). The challenges hindering entrepreneurship in Nigeria have been extensively researched, and there is a general agreement among scholars that several factors hinder students from venturing into different vocational activities. These factors include poor entrepreneurial culture, limited access to funds, a weak knowledge-based economy with low competitiveness, insufficient entrepreneurial education resources, a lack of practical entrepreneurship programs in school curricula, negative societal attitudes towards technical and vocational education, inadequate teaching and learning facilities, government insensitivity towards enterprise creation and expansion strategies, ineffective program management in certain areas, inadequate parental care, erosion of family values and discipline, and political manipulation of youth organisations (Babatunde et al., 2021).

Nevertheless, previous studies have not adequately paid attention to the processes involved in tapping palm wine or traditional alcohol distillation. Although various methods are used across contexts, the non-documentation of these processes in the literature leaves a gap that should be filled. Traditional methods of palm wine and alcohol production need to be documented to strengthen entrepreneurial opportunities and sustain such practices from extinction. Bridging these gaps, the current study was designed to highlight the processes involved in the traditional production of alcohol and the challenges rural producers face.

CONTEXT OF YALA-OBUBRA COMMUNITY

Yala-Obubra community, also known as Nkum Okpambe, is in Obubra Local Government Area of Cross River State, Nigeria. Nkum Okpambe, a Yala tribe of the larger Idoma nation predominantly known for hunting, fishing and crop farming, are extremely peaceful, loving, hospitable and enterprising. The community's people represent a diplomatic melting point between their fellow Obubra and Ikom neighbours. The community is bounded to the north by the cross river, south by the Ofumbongha community, east by the Okosora community and West by the Obokpa community, all in Cross River State.

The community is rich in cultural practices and celebrates different festivals, such as the new yam festival and Okpambe fishing festival, with two massive lakes under their belt. There are numerous stone monoliths, including Lechor Olikplikpo, a mysterious stone that measures just arm's length but comparatively longer than anyone that dares to measure their height with it. However, at the time of writing, the community lacked government presence and social amenities such as motorable roads and electricity. Consequently, most of its indigenes pass through what may be regarded as "hell" to access urban areas. Mostly, the people are farmers, palm wine tappers, fishermen, hunters and artisans. Farmers in this community cultivate cash crops like rice, oil palm, corn, oranges, plantain and cocoa, and food crops like yam, cocoa yam, vegetables and so forth. The population of inhabitants of this community is approximately 3000; however, due to rural-urban migration, half of this population is always found in the community (Independent National Electoral Commission, 2022).

Regarding tourism potential, the community has beautiful sand beaches, a peaceful and serene environment, and a rich cultural heritage. The community also has over 500 natural and artificial fishponds in major lakes. This community was chosen because some of the most experienced palm wine producers and alcohol distillers can be found there. Secondly, the alcohol produced from this part of the country is adjudged the best against the ethanol produced in other regions.

RESEARCH METHODOLOGY

This study adopted qualitative research methods based on the philosophy of interpretivism. Interpretivism is a research philosophy that emphasises the importance of understanding human behaviour and social phenomena from the perspective of the individuals involved (Alharahsheh & Pius, 2020). The interpretive philosophy was considered appropriate for this study because it enabled the researchers to understand the social and cultural significance of palm wine production and alcohol distillation activities from the perspective of the community members involved. The researchers followed the ethnographic case study design in this study. "Ethnographic case studies employ ethnographic methods and focus on building arguments about the cultural, group, or community formation or examining other sociocultural phenomena" (Schwandt & Gates, 2018: 344).

Study Participants

The participants of this study were 10 palm wine tappers (producers) and alcohol distillers in Yala-Obubra, a rural community in Obubra Local Government Area of Cross River State, Nigeria. These participants were carefully chosen due to their years of experience in the activities. The eligibility criteria were that a potential participant is a resident of the Yala-Obubra community, producing palm wine and alcohol, with over 10 years of experience. This cut-off in the age range was to recruit individuals with at least a decade of experience. Participants who are residents of the community that produce only palm wine without alcohol, regardless of their years of experience, were excluded. It is impossible to find rural alcohol producers that do not tap palm wine.

Data Collection and Analysis

The researchers assembled the 10 participants in a hall with chairs arranged semi-circularly. This arrangement enabled the participants to see each other and engage in eye contact and nonverbal communication. After all the participants were seated, the researchers introduced themselves and the purpose of the study thereof. They also explained the nature of the focus group discussion and obtained the participants' informed consent to participate in the study. The researchers began the focus group discussion with an icebreaker activity to help the participants feel comfortable and encourage open discussion. The researchers took notes using notebooks during the discussion to record key points, themes, and ideas, whereas audio recordings of the entire conversation were made through sound recorders. The researchers moderated the entire process by asking follow-up questions, clarifying participants' responses, and encouraging further discussion. Upon completion, the researchers thanked the participants for their time and input and provided an opportunity for them to ask any questions they had. The researchers transcribed the audio recording of the focus group discussion and analysed the data. The researchers used the information from the discussion to identify key themes to develop insights and draw conclusions.

Verifiability and Trustworthiness

Note-taking is a valuable process in qualitative research, as it allows the researcher to capture important points that may not be captured by audio or video recordings. This process allows the researcher to record data in real-time, enabling them to capture a comprehensive discussion account. While note-taking is an important instrument for data collection in qualitative research, there is no need for formal validity and reliability tests, as note-taking is a subjective process that depends on the researcher's interpretation and understanding of the discussion (Hussein, 2022). In the current study, the researchers enhanced validity by ensuring that the notes were taken by one of the co-authors, who was knowledgeable about palm wine production and alcohol distillation and the participants' culture. This enabled the researchers to record the nuances of the discussion and avoid misinterpretation accurately.

Ethical Considerations

The researchers obtained informed consent from all participants before conducting the focus group discussion and recording any audio. The researchers ensured that the participants knew their participation was voluntary and that they could withdraw at any time without consequence. During the focus group discussion, the researchers ensured that all participants were given equal opportunities to express their opinions and that no one was discriminated against based on their tribe, religion, or other personal characteristics. The researchers also ensured that the data collected was used only for the study and was not shared with any third-party individuals or organisations without the explicit consent of the participants. Lastly, the researchers acknowledged the participants' contributions to the study by allowing them to review the findings and provide feedback.

RESULTS OF THE STUDY

The results from the focus group discussion were organised under the following themes in line with the objectives of this study.

Demographic Information

Regarding the demographic characteristics of the respondents, all the participants were males, as the activities are exclusively male-dominated, with an average age of 46.9 years and an average experience of 15.8 years. In terms of education, 30% (n = 3) had no formal education; 20% (n = 2) completed primary education; 40% (n = 4) completed secondary education, whereas 10% (n = 1) completed university education.

Indigenous Palm Wine Tapping Processes in Yala-Obubra

The participants were asked to discuss the steps involved in tapping palm wine according to their traditional methods. The various responses of the participants were synthesised. According to the respondents, the steps involved in palm wine production are:

Selection of mature palm trees: According to the participants, this process involves looking for suitable palm species, checking for mature trees with good health and suitability, and planning for sustainability by avoiding tapping trees that are too young. The tapper goes into the farmland (field) and selects or picks out all the mature palm trees ready for felling based on the numbers or quantities desired for that tapping season. This process is called *ikoro olala.*

Clearing the surrounding area: This process involves using a machete by the tapper to eliminate all grasses, shrubs or other materials that might hinder the next process. Thus, the tapper ensures that the area around the tree is cleared of any obstacles that may hinder the felling process. This process is known and called *Ebeh Onghongho Sigah Ikiro*.

Felling of the palm trees: The third stage is called *Ikiro Ofufu*. It requires falling axes, knives, and hoes for digging. In this stage, the tapper cuts the base of the palm tree using a sharp machete or axe, ensuring that the cut is deep enough to allow the tree to fall in the desired direction. The tapper must ensure that he digs in a manner that allows for the palm tree to fall in the desired direction. In some cases, once the cut is made, a rope could be used to pull the tree in the desired direction to ensure that it falls safely. These could be done by hired or self labour.

Pruning: This is called *Ikiro Opraprah*. The process involves using a sharp machete to cut the fronds and expose the tapping points. The tapper should ensure not to cut too much of the frond to avoid damaging the tree. Thus, the palm fronds are cut out, kept clean, and ready for tapping. Again, this process requires hiring or self labour.

Clearing/cleaning of the road (tracts): In this stage, the tapper keeps the tracts leading to the various felled palm trees clean. This is done to avoid disturbances during tapping and movement from one point to another. The process involves using sharp machetes and can involve both self and employed labour. This process is called *okpolo ongbhogho.*

Heating the cut edges of the pruned palm tree: This step involves the use of prepared bundles of dry palm rods with fire at one end to activate the fresh liquid (palm wine) basically for effective (maximal) production, good taste and prevention of palm-weevils (pests) attack and invasion that can ruin the production process. The fire on the dry palm rods is pointed around the trunk (*Okor*) of the pruned palm tree. The cylindrical trunk represents what is left after pruning all the palm fronds from their crown shafts. This is where the clay pot (*Ukpasi ikiro*) or polythene bag (*waterproof*) is suspended for the collection of palm wine.

Tapping proper: This is the process wherein the tapper cuts off the fermented surfaces of the pruned edge of the palm tree's trunk. Once this is done, clay pots or polythene bags are suspended from collecting all the saps dropping from the palm tree. In Yala-Obubra, this is done two times daily (morning and evening). Tapping requires the use of a very sharp knife to enhance the cutting process and make the surface of the edge of the palm tree trunk to be very clean. In using the machete during this stage, the tapper should not strike the knife; instead, he should use the levering process. Levering involves using the machete as a lever to force it into the topmost edge of the trunk. To do this, the machete is placed against the desired position on the edge of the pruned palm tree trunk, and then pressure is applied to the handle and top region of the knife to push the blade into the palm tree.

Palm wine collection: This stage requires using clay pots (*Ukpasi ikiro*)/polythene bags to collect extracted saps. In this stage, the trimmed edges of the pruned palm are covered with clay pots or polythene bags (depending on what is available or preferable). The clay pots or polythene bags serve as the mechanism where the tapper collects the fresh liquid (palm wine). As a note, the participants discussed that the choice between the use of clay pots or polythene bags depends on several factors such as availability, cost, the quantity of production and the demand of the end-users or consumers of the palm wine.

For availability, they explained that clay pots are sometimes inaccessible due to their very low production in the Yala-Obubra community. They added that a clay pot is sold at the rate of N500, making it very expensive compared to polythene bags sold at the rate of N50. The disparity in the price also affects the preference of most tappers who opt for polythene. They also advised that clay pots are recommended for small-scale production and the use of polythene for large-scale production. During cultural events such as festivals (*Ogbuk*), naming ceremonies (*liyi ojoje*), child dedications (*oyi oyigiyigi*), or hirings (*leligha*), special requests can be made for palm wine collected through clay pots; however, the price is usually different from that collected with polythene bags. In terms of quality, all the participants rated palm wine produced using clay pots to be of higher quality than that produced using polythene bags.

Regarding the quality between the two palm-wine collection methods, some study participants revealed the following:

Participant 4: *The difference between waterproof wine and the one from clay pot (Ukpasi Ikoro) is very clear. The wine from a clay pot is always very cool and fresh, whereas waterproof wine is always very hot. When you drink the two, you will taste the difference without anyone telling you.*

Participant 8: *Palm wine collected through clay pots has a wider acceptance among consumers than those collected from waterproofs. Most drinkers complain of stomach aches after consuming palm wine tapped using waterproof. Because of this, anytime I use waterproof to tap, I struggle to sell up to 40 litres of palm wine in a day. However, whenever I use clay pots, people rush to buy them.*

Participant 3: *Because of the low acceptance of waterproof collected palm wine, I do not make commercial sales of fresh palm wine whenever I use waterproof. I only supply fresh palm wine when not producing in large quantities. Usually, when I produce in large quantities, like most other tappers here, I aim to convert fresh palm wine to alcohol after fermentation. Waterproofing is preferred in this case* *due to the cost associated with having to buy many clay pots that are very delicate. Nevertheless, when the number of palm trees I felled is like 10 or anything less, I use my clay pots, and during that time, fermentation to alcohol is not done; I only supply palm wine sellers or people with occasions or based on special requests.*

Causes of the Poor Taste of Palm Wine from Some Tappers

Respondents were asked to discuss why some palm wine products yield a poor or sour taste. After deliberations among the participants, the following factors were noted as causes of poor palm wine taste. These included individual differences in tapping approach, poor maintenance culture of the tapping instrument such as knife, sharpening, washing (cleaning) of the pots, invasion of certain insects like Agàyankpa, Edangblan and bees, tapping period (intervals), i.e., tapping two times daily (morning and evening) at inappropriate time schedules.

Reasons for Differential Drunkenness Patterns among Palm Wine Consumers

Participants were further asked to explain why some palm wines are more potent in promoting drunkenness among individuals consuming the same quantity. According to the participants, the toxicity of palm wine is based on the following factors: the soil texture, the age (maturity) of palm trees, the tapping season (rainy or dry season, but the latter is recommended) and the tapping apparatus (e.g., polythene bags, clay pots). The participants also mentioned personal factors such as blood group, the nature (ability) of the individual drinking the palm wine and family traits. The researchers visited the palm wine production site of Participant 5, aged 64 years, with over 20 years of experience in palm wine production and alcohol distillation. Subsequently, the following photographs were taken to substantiate the writing.

Indigenous Alcohol Distillation Practices in Yala-Obubra

There are a series of tapping processes that lead to the extraction of alcohol in Yala-Obubra, as revealed by this study's participants. The processes which were thoroughly discussed include:

Collection of palm wine: This is the process whereby the "tapped" fresh palm wine is collected to a central point to prepare for the distillation process. It requires several containers, such as pots, drums (both plastics/tin), and gallons of different sizes. Figures 5 and 6 show how palm wine has been collected into different containers.

Fermentation: the fresh palm wine is stored in the container for about 1-2 weeks before the distillation processes based on the boast of fresh palm collected daily by the tapper. Fermentation duration lasts between 1-3 weeks for proper and good alcohol. The tapper learns that the stored wine is ready for distilling by observation and the amount of yeast that settles at the bottom of the stored containers. Figure 7 shows palm wine undergoing fermentation at a production site.

Materials required: Several apparatus/equipment is needed here, which are locally produced or made available by the tapper. These include boiling pots/drums, water, firewood, distilling pipes, receivers, funnels, corks, crushed cassava or rotten plantain/tissue or soak gari, carved wooden boats, sharp pointers, and stone of big sizes. The pots/boiling drums serve as an instrument for heating the palm wine to change of stage, i.e., liquid-gas. Fire/firewood provides the heating effects for cooking/boiling procedures. The cork connects the distilling pipes to the pots or boiling drums. A carved wooden boat

Figure 1. Front view of a pruned palm tree producing fresh palm wine

Figure 2. Back view of pruned palm tree producing fresh palm wine

serves as the condenser that aids change of state (that is, palm-gas-alcohol). Crushed cassava, gari or rotten plantain tissue serve as seals that prevent evaporation (escape of gaseous alcohol to the atmosphere. Stones serve as locally made triple stands where the boil pods or boiling drums are placed. Dry

Figure 3. Pruned palm tree producing wine with cleaned Rafias on top of it

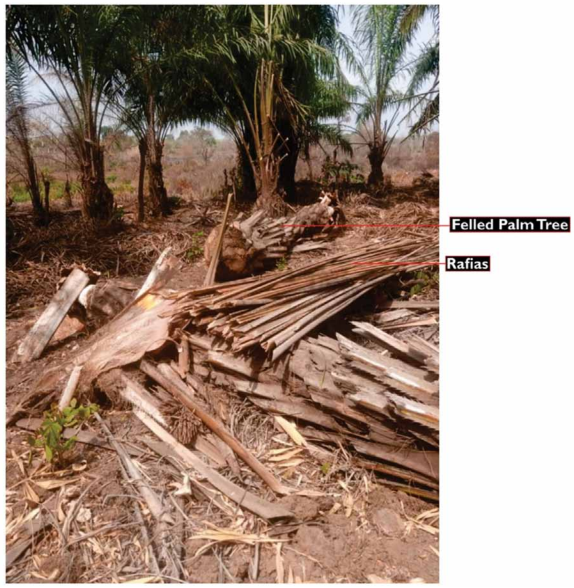


Figure 4. Pruned palm tree producing palm wine fitted with polythene bag

woods serve as sources of fuel for heating that aid temperature change. Receivers are for the collection of distilled spirits after condensation. In the olden days, empty bottles, calabash, and pots were used; however, gallons are common nowadays. The funnel prevents waste during high wind or breeze and aids the safe collection of alcohol. The pointer helps to direct the flow of alcohol into the receiver, whereas the cotton wool is used for filtration.

Figure 5. Palm wine collected and stored in containers

Figure 6. Collected palm wine undergoing fermentation

Figure 7. Left view of palm wine undergoing heating

Alcohol Distillation: The distillation of palm wine typically involves using a pot still, a simple device consisting of a large pot with a lid, a tube or pipe to carry the vapour, and a condenser (a canoe-like object) to cool and collect the distilled liquid. The pot is filled with fermented palm wine and heated at a regulated temperature over a fire. As the liquid is heated, the alcohol and other volatile compounds evaporate and rise through the tube or pipe. The vapour is then directed through the condenser, which cools the vapour and causes it to condense back into a liquid. The resulting liquid is collected in a container and can be further distilled to increase its alcohol content. The process can be repeated several times until the desired alcohol concentration is reached. However, it is important to note that the traditional distillation process for palm wine can be risky and potentially dangerous. This is because impurities can be present in fermented palm wine, producing toxic compounds like methanol. Additionally, using open flames and other heat sources can pose a fire hazard if not used properly. To mitigate these risks, it is recommended that only experienced individuals carry out the distillation process and that proper safety measures are taken, such as using a well-ventilated area and ensuring that the steel is free from contaminants. Figures 8 to 12 display some of these procedures.

Challenges to Traditional Palm Wine Production and Distillation of Spirits In Yala-Obubra

The participants identified the following challenges in their palm wine production and alcohol distillation practices.

Declining palm tree population: One of the major challenges to traditional palm wine production is the declining population of palm trees. Palm trees are the primary source of palm wine, and as the number of trees decreases, the availability of palm wine also decreases. This is especially so because, in the Yala-Obubra community, the destructive approach to palm wine tapping is prevalent. This means that as a palm tree is felled and wine is tapped off it, that tree ceases to exist. The recent surge in palm oil prices also challenges indigenous palm wine tappers. Buttressing this point, participant 2 said:

One of our biggest problems right now is having access to land with plenty of mature palm trees. God planted most of the palm trees used for tapping, and we were only born to meet and harvest them. Some of us, like me, have finished tapping all the mature palm trees in our family forest; I now buy from community members willing to sell.

Moreover, participant 9 added:

Recently, there has been a massive increase in the price of palm oil. As a result, many people are no longer selling their palm trees for wine production. They prefer to harvest palm fruits and process them into oil which they sell at very good prices. Palm kernel is now a hot cake; almost all oil producers sell palm oil and its associated kernel. This gives them extra income, and by comparison, it seems rational to produce palm oil rather than palm wine which destroys the trees.

Climate change: Climate change has also been identified as a challenge to traditional palm wine production in the Yala-Obubra community. Changes in temperature and rainfall patterns can affect the production of sap and the growth of palm trees, leading to reduced yields. For instance, participant 7 favoured the dry season as the best time for palm wine production, with the highest yield. Participant 7 added that:

During the dry season, the wine comes out very well when taping. You will tap a few palm trees and have production in relatively large quantities, but during the rainy season, palm wine is expensive because of low production. Only a few individuals tap during that time.

High cost of production: The high cost of production is another challenge to traditional palm wine tappers. The equipment and materials needed for palm wine production and alcohol distillation are expensive, making it difficult for small-scale producers to compete with larger commercial producers. Most participants, who do not have palm trees on their private or family lands, attested to buying them at a high cost from community members willing to sell. Participant 6 felt the following regarding the high cost of production:

Before now, we used to buy litre 20 at the rate of N350 and litre 25 at the rate of N500, but today, if you go to our market, you cannot afford them any longer. An empty litre 20 is now sold for N1800, while 25 litres is sold at N2000 or N2200. In a tapping season, you may need ten of these gallons. Then, for alcohol production, you need big rubbers (Baba Sala or Buta) for the fermentation of palm wine. The of these depend on the size of the rubber, but the smallest and cheapest one out there is not less than N14,000 Naira, and you may need about four to five depending on the number of palm trees felled. These rubbers when my father used to tap, we used to buy these for less than N10, but Nigeria has changed.

Lack of access to credit: Small-scale producers also faced challenges accessing credit to invest in their operations. This limits their ability to purchase palm trees, equipment and materials needed for production and ultimately limits their ability to grow their businesses. Study participants also complained about a lack of government support to expand their businesses. Most participants (n = 6) are unaware of the possibility of accessing credit facilities. The lack of awareness and knowledge about financial support opportunities led to traditional alcohol distillers not taking advantage of credit opportunities. Without access to credit, traditional alcohol distillers could not invest in their businesses, leading to low productivity and limited income. This, in turn, made it difficult for them to repay debts and enjoy a better living amidst family responsibilities. Instead, some participants (n = 5) indicated that they borrow monies from informal credit facilities such as moneylenders, relatives, *Osusu*, age grades and other community-based committees to finance their businesses at a very high (50%) interest rate.

Lack of social amenities: Lack of social amenities was discussed as one of the most threatening challenges faced by traditional palm wine producers and alcohol distillers. Social amenities refer to basic public facilities and services such as electricity, clean water, sanitation, and transportation. When collecting this data, the Yala-Obubra community had no motorable road or means of transportation and electricity. The citizens only generated electrical power through privately-owned generators. Since transportation is crucial for producers to transport raw materials and finished products to markets, this presented a big threat to them. Some respondents discussed how they could not visit neighbouring and distant markets to sell their products. Participant 8 said:

For us here in Yala-Obubra, the road is our biggest problem. Every year, politicians come here to campaign that we vote for them, but when they emerge victorious, we only see them again after four years in the next election season. Because of no motorable roads, many buyers of palm wine or alcohol do not enter our village. The few that manage to enter our community take monopolistic and sometimes oligopolistic advantage of the people by offering very low prices for our products. Since we do not have a choice, we sell our products at whatever fees they are willing to pay. It is quite understandable because the lack of a good road network to our community makes buyers spend so much on transportation to get here.

Participant 5 added that:

We suffer here a lot due to the lack of roads and electricity. As my brother has already said, buyers from outside treat us as they like since they know we have no other options but to do their bidding. Another big problem for us here is no light. I have heard from some of my friends that palm wine can be boiled and refrigerated to maintain freshness; we cannot practice such here because there is no light to power our fridges. Do we even buy fridges in this community? No one buys such because it will be a waste of money.

Stealing: Traditional alcohol production often involves using valuable raw materials and equipment, which can make them a target for theft. The participants indicated that theft could occur at various stages of the production process, from collecting raw palm wine materials to the final distillation of alcohol. They said the theft could occur at the production site or home during storage. For example, the participants generally agreed that community members sometimes visit their site to steal fresh palm for drinking. Other times, they target other resources such as pipes, fermenting rubbers, etc. Participant 4 emphatically stated that:

Stealing happens to us almost every season, from fellow tappers to community members. For example, there was a time when a tapper from this community stole my distillation pipe. This distillation pipe was rated the best in town due to its thickness and ability to promote condensation with a limited wa-

ter supply. Many tappers considered me lucky to own such a pipe; it was inherited from my father, my friend is here now, and he can testify. We searched the entire place and visited all alcohol production sites looking for the pipe. Fortunately, we found the pipe in the hand of a tapper who claimed to have bought it from an undisclosed man.

DISCUSSION OF FINDINGS

The traditional process of tapping palm wine in Yala-Obubra involves selecting mature palm trees, clearing the area, felling the trees, pruning the fronds, clearing the tracts, heating the cut edges of the trunk, tapping the fermented surfaces, and collecting the sap in clay pots or polythene bags. This finding aligns with the results obtained by Mba et al. (2019), which also revealed that the traditional tapping process employed by palm wine tappers involves cutting open or felling palm trees to collect the sap for fermentation. The result also supports Apeh and Opata's (2019) finding that in Igboland, particularly in Enugu State, three methods of palm wine production are used -inflorescence, bud and felling methods. Under the felling method (like the approach practised by Yala-Obubra tappers), the cited researchers explained that the oil palm tree is either cut down or uprooted and tapped at the bud. In the case of a felled tree, the leaves are removed to expose the inner part of the bud, which is then left for two to four weeks. Subsequently, a chamber with dimensions of approximately 15 cm x 15 cm is created on the bud and covered with leaves to shield it from direct sunlight, which may cause drying. After an additional three days, a hole is drilled through the chamber below the bud, where a funnel is attached. The wine seeps into the chamber, flows through the hole, collects in the funnel, and fills the gourd or pot beneath the bud.

The current study further discovered that the choice between clay pots and polythene bags as methods of palm wine collection depends on availability and cost, with clay pots being preferred for small-scale production. Palm wine collected with clay pots is considered higher quality and more widely accepted by consumers. This finding aligns with the position of Apeh and Opata (2019) that the quality of palm wine could be affected by production methods. However, the authors revealed, through a comparison of three production methods, that the inflorescence method was superior, followed by the bud and felling methods. The authors added that proficiency and skill are essential for effectively executing each of these techniques, as they demand significant training and dexterity. This finding has implications for small-scale producers who can prioritise using clay pots for palm wine collection to meet consumer preferences and increase their market share.

The poor taste of palm wine from some tappers can be attributed to various factors identified by the participants. These factors include differences in tapping approaches among individuals, inadequate maintenance of tapping instruments such as knives, lack of proper cleaning of pots, infestation by insects like Agàyankpa, Edangblan, and bees, and tapping at inappropriate intervals or time schedules. These findings highlight the importance of proper tapping practices, instrument maintenance, and hygiene measures to produce high-quality palm wine with desirable taste characteristics. Addressing these factors can improve palm wine quality and consumer satisfaction in the community.

According to the participants, the differential patterns of drunkenness among palm wine consumers can be attributed to various factors. These factors include the soil texture, age (maturity) of palm trees, tapping season (preferably dry season), tapping apparatus (polythene bags or clay pots), personal factors such as blood group, individual's nature or ability to handle alcohol and family traits. This finding varies from the results obtained by Mbadiwe et al. (2021), showing that the gender and age of consumers were

Figure 8. Right view of palm wine undergoing heating

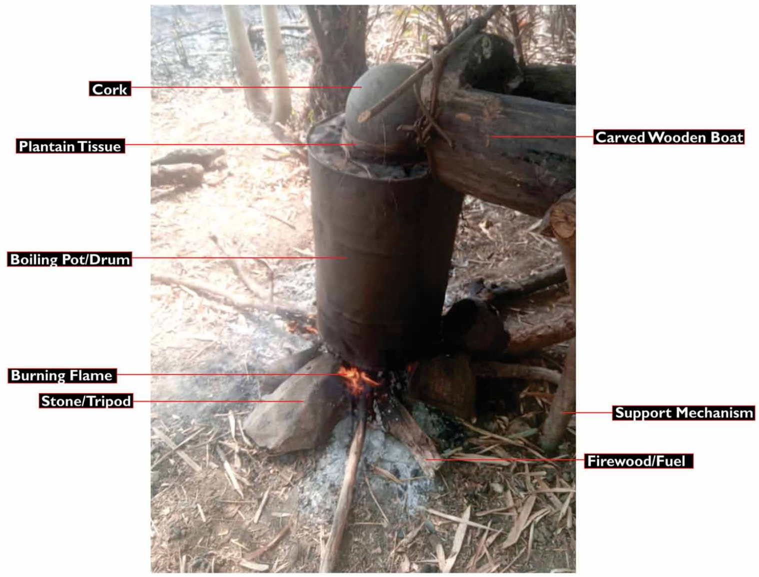
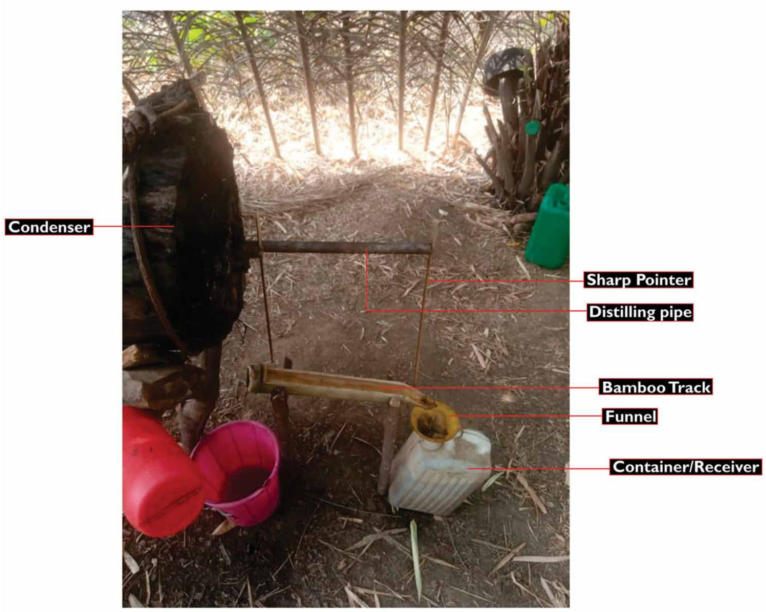


Figure 9. A canoe-like device used for condensation

associated with differential alcohol patterns among youths. The variation in the results of the cited and the present study is understandable since the former focused on factors specific to consumers. In contrast, the current study focused on producer-related factors responsible for differences in drunkenness patterns among consumers. Thus, understanding these factors can help consumers make informed choices, and tappers produce palm wine with desired characteristics.

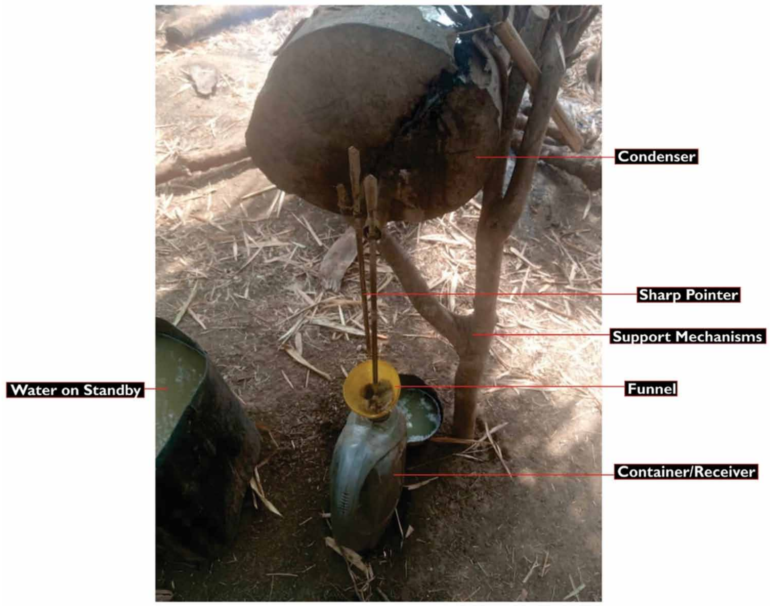
Figure 10. Top view of the condenser with water inside for cooling the pipes

Figure 11. Alcohol collection point



The indigenous alcohol distillation practices in Yala-Obubra involve a series of processes. These include the collection of palm wine into various containers, fermentation of the palm wine for 1-2 weeks, and using locally made apparatus such as boiling pots, distilling pipes, and receivers. The alcohol distillation process utilises a pot still, where fermented palm wine is heated, causing the alcohol and volatile

Figure 12. Closer view of the alcohol collection process



compounds to evaporate and then condense back into a liquid using a condenser. However, it is important to note that traditional distillation methods can be risky, and precautions should be taken to ensure safety. These processes are at variance with the ones reported by Hue et al. (2022) in Vietnam, where alcohol is produced using glutinous rice and indigenous herbs. The author reported that rice and specific forest leaves are used to culture wine starters known as "banh men la" by the local community. After alcoholic fermentation and distillation, the product is white wine or "ruou trang." These processes also misalign with those reported by Asrani et al. (2019) in India, where people use medicinal plants to produce alcoholic drinks because these plants have special properties. In addition to medicinal plants, people use fruits, vegetables, grains, and herbs to make these beverages. One similarity between all the studies cited and the current one is that fermentation played a key role in alcohol production. Nevertheless, the indigenous alcohol distillation practices in Yala-Obubra hold cultural and traditional significance, reflecting the community's rich heritage and resourcefulness. These practices involve using locally made apparatus and highlight the community's adaptability. However, it is important to prioritise safety measures and promote education to prevent accidents and ensure the well-being of the distillers.

In Yala-Obubra, traditional palm wine production and alcohol distillation face several challenges. These include a declining palm tree population due to destructive tapping practices and the preference for palm oil production, the impact of climate change on palm wine yields, the high cost of production and lack of access to credit for small-scale producers, the absence of social amenities such as electricity and transportation, and the issue of theft at various stages of the production process. These challenges jeopardise the sustainability of traditional practices and call for interventions to support the producers and preserve their cultural heritage. The challenges faced by traditional palm wine producers and alcohol distillers in Yala-Obubra have multiple underlying reasons. The declining palm tree population results from destructive tapping practices and the shift towards palm oil production, driven by economic factors. This agrees with the result of Mba et al. (2019) that the practice of palm wine tapping results in many palm trees becoming unproductive or dying, and there is a lack of emphasis on tree planting, further exacerbating the negative impact on the palm tree population. This study discovered that climate change contributes to the situation by affecting palm wine yields. The result also agrees with the results of Nwashindu and Onu (2019) that the risk of death associated with climbing tall palm trees, environmental factors such as cold or swampy plantations, malaria sickness of the tappers and attacks from wild animals are other challenges affecting palm wine production. Other challenges identified in the current study include the high cost of production, lack of access to credit, absence of social amenities, and incidents of theft pose additional threats. Addressing these challenges is crucial to ensure the sustainability of traditional practices, support the livelihoods of producers, and preserve the cultural heritage of Yala-Obubra.

IMPLICATIONS OF THE FINDINGS FOR STUDENTS' ENTREPRENEURSHIP DEVELOPMENT

Palm wine production and alcohol distillation are economic opportunities that can foster entrepreneurship development and drive sustainable economic growth. The two economic activities have been practised for centuries in many rural communities worldwide, and they offer significant opportunities for income generation, employment creation, value addition, and innovation. The importance of palm wine production and alcohol distillation for entrepreneurship development is that they offer opportunities for students to learn practical skills and gain hands-on experience in various areas, including agriculture, food processing, chemistry, and business management. For example, producing palm wine requires knowing how to tap and collect sap from palm trees, how to ferment and age the sap to create the desired flavour and alcohol content, and how to package and market the finished product. Similarly, alcohol distillation involves understanding the chemical processes involved in separating and concentrating ethanol and how to design and operate distillation equipment, manage fermentation processes, and comply with regulatory requirements.

In addition to these technical skills, palm wine production and alcohol distillation also require entrepreneurial skills such as marketing, sales, leadership and financial management. Students who engage in these activities must learn to identify and target potential customers, develop effective marketing strategies, negotiate with suppliers and distributors, manage cash flow and inventory, and comply with legal and regulatory requirements. By gaining experience in these areas, students can develop valuable skills and knowledge that will serve them well in many career paths.

Another reason palm wine production and alcohol distillation are important for entrepreneurship development is that they can provide income and economic opportunities for students and their communities. According to Apeh and Opata (2019), palm wine production is a significant means of generating income and employment opportunities for rural farmers. In many regions where these practices are common, they have long been important sources of income and employment, particularly for small-scale producers and entrepreneurs. By learning how to produce and sell these products, students can create their income streams and contribute to the economic development of their communities. Moreover, they can leverage technology and social media platforms to reach a wider market and increase their income. Many producers and alcohol distillers explained that the business has a huge income generation and recommend that people without better-paying jobs engage in them. Some tappers have trained their wives, children and other dependents up to the university level using proceeds from these two indigenous practices. Others have built houses, bought cars and motorcycles, and recorded other achievements as palm wine tappers. Therefore, these two practices can generate income for students, especially unemployed graduates, as they wait for better employment opportunities.

CONCLUSION AND RECOMMENDATIONS

This study used a qualitative approach to investigate and describe palm wine production and alcohol distillation. The study's findings revealed several steps involved in producing palm wine and distilling alcohol. Palm wine tappers and alcohol distillers in the Yala-Obubra community face various challenges in the production process. Palm wine production and alcohol distillation are important economic activities with good entrepreneurship potential for students and graduates. Therefore, higher education students and graduates can consider these as alternative sources of income generation and poverty eradication rather than rely solely on white-collar jobs. The study underscores the importance of preserving and promoting traditional practices in Nigeria. The traditional practices of palm wine production and alcohol distillation offer significant socio-economic benefits that could contribute to the country's development. The following recommendations were made to help address the challenges facing palm wine producers and alcohol distillers in the Yala-Obubra community.

- Tappers in Yala-Obubra need to adopt sustainable methods of palm tree management that promote the longevity of the trees. Since palm oil production was discussed as the reason for the lack of access to palm trees for tapping, destructive approaches to palm wine production should be discouraged. This can include tree pruning instead of felling, controlled tapping methods that allow trees to recover and planting new palm trees to increase the population.
- The impact of climate change can be mitigated by adopting climate adaptation strategies such as planting drought-tolerant palm tree varieties, implementing rainwater harvesting, and using irrigation to ensure adequate water supply during dry seasons.
- The government and other organisations can financially support traditional palm wine producers and distillers by offering low-interest loans and grants to purchase equipment and materials, expand their businesses, and improve their livelihoods.
- Providing social amenities such as motorable roads, electricity, clean water, sanitation, and transportation will greatly benefit traditional palm wine producers and distillers in the Yala-Obubra community.
- Alcohol producers can use traditional fencing around the production site or hire a security guard to monitor the site. This can help producers minimise theft risk and protect their valuable raw materials and equipment.

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KEY TERMS AND DEFINITIONS

Distilled Spirits: These, also known as hard liquor or alcoholic beverages, are produced by distilling fermented grains, fruits, or vegetables. Examples include whiskey, vodka, gin, rum and brandy, which typically have higher alcohol content than beer or wine.

Entrepreneurship: Entrepreneurship refers to designing, launching, and running a new business venture in the face of uncertainty and risk to create value, generate profits and achieve sustainability. Entrepreneurship drives economic growth, creates jobs, fosters innovation, and addresses social challenges.

Fermentation: The process by which microorganisms such as yeast break down sugars into alcohol, carbon dioxide, and other byproducts. In palm wine production, fermentation converts sap from palm trees into an alcoholic beverage.

Indigenous Practices: This refers to traditional knowledge, skills, and techniques passed down through generations in a particular community or culture. These practices are often closely tied to the local environment and resources and are shaped by social and cultural factors.

Palm Wine Production: This refers to the process of extracting sap from the trunk of palm trees, which is then fermented to produce a beverage with low alcohol content. This practice is common in many parts of Africa and plays an important cultural and economic role in many communities.